

Kernel, Exit, and Occupation Laws of a Skew Brownian Interface

Route-A source-local certificate HCS-C266

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Abstract

We give one convention-complete theorem for zero-drift skew Brownian motion. Symmetric local time is frozen explicitly; the Lebesgue kernel and its jump, the speed-measure symmetric kernel, complete resolvent, every two-sided exit probability, both discounted exit transforms, mean exit time, and generalized arcsine occupation law are derived. Ordinary Brownian motion and both one-sided reflected limits remain in the atlas. Executable checks separate finite regression from proof and enforce a strict Route-A stopping boundary.

1 Frozen interface and transition law

Let B be standard Brownian motion and let $L^0(X)$ denote *symmetric semimartingale local time*. For $p \in [0, 1]$, put $\theta = 2p - 1$ and freeze

$$X_t = x + B_t + \theta L_t^0(X). \quad (1)$$

Harrison and Shepp proved strong existence and pathwise uniqueness exactly for $|\theta| \leq 1$ and identified the solution by independently making each reflected Brownian excursion positive with probability p [1]. Thus (1) fixes a normalization; replacing symmetric by right local time without changing the coefficient defines a different equation.

Write $\phi_t(z) = (2\pi t)^{-1/2} e^{-z^2/(2t)}$.

Theorem 1 (Lebesgue kernel and generator). *For $t > 0$, the transition density with respect to Lebesgue measure is*

$$q_p(t; x, y) = \phi_t(y - x) + (2p - 1) \operatorname{sgn}(y) \phi_t(|x| + |y|). \quad (2)$$

It is nonnegative, conservative, and satisfies Chapman–Kolmogorov. On the standard Feller core, $Af = f''/2$ off zero and

$$pf'(0+) = (1 - p)f'(0-). \quad (3)$$

Proof. In each open quadrant, (2) is a sum of Gaussian images and solves the heat equation. As a function of the starting point it is continuous at zero; the two derivatives obey (3). Split integration at zero. Half-Gaussian integration gives mass one and positivity, while the same split followed by the Gaussian convolution identity gives Chapman–Kolmogorov. Hence (2) identifies the Feller semigroup of the excursion-sign solution. $\square$

The one-sided terminal limits satisfy $q_p(t; x, 0+)/q_p(t; x, 0-) = p/(1 - p)$ when $0 < p < 1$. Thus the Lebesgue density is not continuous across the terminal interface unless $p = 1/2$.

2 Speed symmetry and resolvent

For $0 < p < 1$, choose scale and speed densities

$$s'(x) = \begin{cases} (1 - p)^{-1}, & x < 0, \\ p^{-1}, & x > 0, \end{cases} \quad m'(x) = \begin{cases} 2(1 - p), & x < 0, \\ 2p, & x > 0. \end{cases}$$

The scale condition is exactly (3). Although (2) jumps in its terminal Lebesgue coordinate, $k_p(t; x, y) = q_p(t; x, y)/m'(y)$ is symmetric. Indeed, for $x > 0 > y$ the forward and reversed Lebesgue densities are $2(1 - p)\phi_t(|x| + |y|)$ and $2p\phi_t(|x| + |y|)$. Same-side symmetry is immediate. The semigroup is therefore self-adjoint and contractive on $L^2(m)$.

Theorem 2 (complete resolvent). *For $\lambda > 0$ and $k = \sqrt{2\lambda}$,*

$$r_\lambda(x, y) = \int_0^\infty e^{-\lambda t} q_p(t; x, y) dt = \frac{e^{-k|x-y|} + (2p-1) \operatorname{sgn}(y) e^{-k(|x|+|y|)}}{k}. \quad (4)$$

Proof. Apply $\int_0^\infty e^{-\lambda t} \phi_t(z) dt = e^{-k|z|}/k$ to both images in (2). Formula (4) independently satisfies $(\lambda - A)r = 0$ off the pole and interface, condition (3), and the unit resolvent source normalization: its x -derivative has jump -2 at $x = y$, so $-(1/2)$ times that jump is the unit delta mass. $\square$

3 Every two-sided exit observable

Let $a, b > 0$ and $\tau = \inf\{t \geq 0 : X_t \notin (-a, b)\}$. Boundary labels are retained rather than folded into one transform.

Theorem 3 (hitting probability and mean). *With $D = pa + (1-p)b$,*

$$\mathbb{P}_x(X_\tau = b) = \begin{cases} p(x+a)/D, & -a \leq x \leq 0, \\ [pa + (1-p)x]/D, & 0 \leq x \leq b. \end{cases} \quad (5)$$

Put $D_m = (1-p)b + pa$ and

$$A = \frac{(1-p)(b^2 - a^2)}{D_m}, \quad C = \frac{p(b^2 - a^2)}{D_m}, \quad B = \frac{ab[pb + (1-p)a]}{D_m}.$$

Then

$$\mathbb{E}_x \tau = \begin{cases} -x^2 + Cx + B, & x \leq 0, \\ -x^2 + Ax + B, & x \geq 0. \end{cases} \quad (6)$$

Proof. Equation (5) is the unique piecewise affine solution of $h'' = 0$ with boundary values zero and one, continuity, and (3). Equivalently it is $(s(x) - s(-a))/(s(b) - s(-a))$ using the scale density displayed above (4). Equation (6) is the unique piecewise quadratic solution of $v''/2 = -1$, with zero boundary data and the same two interface conditions. $\square$

Theorem 4 (discounted exit sides). *For $0 < p < 1$, let $\rho = (1-p)/p$, $k = \sqrt{2\lambda}$, and*

$$Q = \cosh(kb) \sinh(ka) + \rho \sinh(kb) \cosh(ka).$$

The right-boundary transform $R_\lambda(x) = \mathbb{E}_x[e^{-\lambda\tau}; X_\tau = b]$ is

$$R_\lambda(x) = \begin{cases} \sinh(k(x+a))/Q, & x \leq 0, \\ [\cosh(kx) \sinh(ka) + \rho \sinh(kx) \cosh(ka)]/Q, & x \geq 0. \end{cases} \quad (7)$$

The left transform is $R_\lambda^{1-p}(-x; b, a)$, and the sum is $\mathbb{E}_x e^{-\lambda\tau}$.

Proof. Solve $u''/2 = \lambda u$ on each half. Propagation of $(u, u'/k)$ uses $M(\ell) = \begin{pmatrix} \cosh(k\ell) & \sinh(k\ell) \\ \sinh(k\ell) & \cosh(k\ell) \end{pmatrix}$. At zero continuity preserves the first coordinate while (3) multiplies the second by ρ . Applying $u(-a) = 0, u(b) = 1$ gives (7). Reflection exchanges the two boundary labels. $\square$

4 Occupation law and endpoint atlas

Start at zero and set $U_t = t^{-1} \int_0^t \mathbf{1}_{\{X_s > 0\}} ds$.

Theorem 5 (generalized arcsine law). *For $0 < p < 1$, U_t has a law independent of $t > 0$, with density*

$$f_p(u) = \frac{p(1-p)}{\pi \sqrt{u(1-u)} [p^2(1-u) + (1-p)^2 u]}, \quad 0 < u < 1. \quad (8)$$

It satisfies $\mathbb{E} U_t = p$. At $p = 0$ and $p = 1$ the law is respectively an atom at zero and one.

Proof. At inverse local time, positive and negative excursion durations are independent stable-1/2 subordinators with Laplace coefficients p and $1 - p$. For iid standard positive stable-1/2 variables S_+, S_- , the ratio is therefore

$$\frac{p^2 S_+}{p^2 S_+ + (1 - p)^2 S_-}.$$

The standard ratio calculation gives (8), the two-ray Lamperti law; this occupation construction is also treated in [2]. The source has two published corrections [3, 4]; no corrected multivariate-density or transform formula is imported here, since (8) follows directly from the stable ratio above. Substitution $u = \sin^2 v$ verifies normalization, and $f_p(u) = f_{1-p}(1 - u)$. Finally Fubini and $\mathbb{P}_0(X_s > 0) = p$ give $\mathbb{E}U_t = p$. $\square$

For $p = 1/2$, the image term in (2) vanishes, the speed measure m is Lebesgue measure, and (8) is the classical arcsine density. At $p = 1$ or 0 , after the first hit of zero all excursions have one sign, producing one-sided reflection. A coefficient $|\theta| > 1$ admits no solution and is not part of the model. Drifted, unequal-diffusivity, and sticky interfaces are not claimed.

5 Executable receipt and Route-A boundary

The certificate stores 135 kernel, 9 speed-symmetry, 18 resolvent, 50 exact exit, 54 discounted-exit, 6 Chapman–Kolmogorov, and 3 occupation rows. A producer-independent implementation closes 963 assertions, SymPy closes 133 generic and exact identities, fresh replay is byte-identical, and 16/16 stale/repaid-hash mutations are rejected. These 275 rows are regression oracles; the preceding proofs hold for all admissible parameters.

This recurrent noncompact diffusion has no isolated primitive source-orbit ledger, rational-prime carrier, logarithmic prime clock, or source Artin–Mazur product. Speed-space self-adjointness is a formal source hint, not a target operator. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the tuple is

$$(\text{AO_FAIL}, \text{A1_FAIL}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_FORMAL_HINT}).$$

The verdict is `ROUTE_A_REJECTED`; Route B is disabled. We claim no target arithmetic local data, Euler factor, root number, automorphy, target divisor, functional equation, target zero match, or Hilbert–Pólya operator.

References

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- [2] T. Appuhamillage, V. Bokil, E. Thomann, E. Waymire, and B. Wood, *Occupation and local times for skew Brownian motion with applications to dispersion across an interface*, Ann. Appl. Probab. **21** (2011), 183–214, [doi:10.1214/10-AAP691](https://doi.org/10.1214/10-AAP691).
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- [4] T. Appuhamillage, V. Bokil, E. Thomann, E. Waymire, and B. Wood, *Second errata to “Occupation and local times for skew Brownian motion with applications to dispersion across an interface”*, Ann. Appl. Probab. **34** (2024), 5842–5844, [doi:10.1214/24-AAP2106](https://doi.org/10.1214/24-AAP2106).